

## Assignment-1

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1. Create a calculator program.

A Calculator class having 4 function (add, multiply, subtract, divide)

Ask to the user for an option to perform the operation.

**Program:-**

```
package day1.Array;
import java.util.Scanner;
import static java.lang.System.exit;

public class Calculator {
    public double add(double num1, double num2){
        return num1+num2;
    }
    public double sub(double num1, double num2){
        return num1-num2;
    }
    public double mul(double num1, double num2){
        return num1*num2;
    }
    public double div(double num1, double num2){
        return num1/num2;
    }
}

public static void main(String []args){
    Scanner sc= new Scanner(System.in);

    double num1,num2;
    Calculator cal= new Calculator();

    do {
        System.out.println("1.Addition");
        System.out.println("2.Subtraction");
        System.out.println("3.Multilication");
        System.out.println("4.Division");
        System.out.println("0.Exit");
        System.out.print("Enter choice: ");
        int choice=sc.nextInt();
```

```

        switch(choice){
            case 1:
                System.out.print("Enter first number: ");
                num1= sc.nextDouble();
                System.out.print("Enter second number: ");
                num2= sc.nextDouble();
                double addition= cal.add(num1,num2);
                System.out.println("Addition: " +
addition);

                break;
            case 2:
                System.out.print("Enter first number: ");
                num1= sc.nextDouble();
                System.out.print("Enter second number: ");
                num2= sc.nextDouble();
                double subtraction= cal.sub(num1,num2);
                System.out.println("Subtraction: " +
subtraction);

                break;
            case 3:
                System.out.print("Enter first number: ");
                num1= sc.nextDouble();
                System.out.print("Enter second number: ");
                num2= sc.nextDouble();
                double multiplication= cal.mul(num1,num2);
                System.out.println("Multiplication: " +
multiplication);

                break;
            case 4:
                System.out.print("Enter first number: ");
                num1= sc.nextDouble();
                System.out.print("Enter second number: ");
                num2= sc.nextDouble();
                double division= cal.div(num1,num2);
                System.out.println("Division: " +
division);

                break;
            case 0:
                System.out.println("Successfully
Exited!!!!");

                exit(0);
            default:
                System.out.println("Invalid
Input.....");

```

```
        }  
        }while(cal!=null);  
        sc.close();  
    }  
}
```

Outputs:-

```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ  
1.Addition  
2.Subtraction  
3.Multilication  
4.Division  
0.Exit  
Enter choice: 1  
Enter first number: 12  
Enter second number: 20  
Addition: 32.0  
1.Addition  
2.Subtraction  
3.Multilication  
4.Division  
0.Exit  
Enter choice: 2  
Enter first number: 33  
Enter second number: 4  
Subtraction: 29.0  
1.Addition  
2.Subtraction  
3.Multilication  
4.Division  
0.Exit  
Enter choice: 3
```

5.Invalid Input

4.Division

0.Exit

Enter choice: 3

Enter first number: 4

Enter second number: 3

Multiplication: 12.0

1.Addition

2.Subtraction

3.Multilication

4.Division

0.Exit

Enter choice: 4

Enter first number: 10

Enter second number: 2

Division: 5.0

1.Addition

2.Subtraction

3.Multilication

4.Division

0.Exit

Enter choice: 6

Invalid Input.....

1.Addition

2.Subtraction

3.Multilication

4.Division

0.Exit

Enter choice: 6

Invalid Input.....

1.Addition

2.Subtraction

3.Multilication

4.Division

0.Exit

Enter choice: 0

Successfully Exited!!!!

2. Create an array using String.  
Then populate array using String.  
Then read the populated String array.

### Program:-

```
package day1.Array;
import java.util.Scanner;

public class StringArray {
    public static void main(String []args){
        Scanner sc= new Scanner(System.in);
        System.out.print("Enter the size of array: ");
        int num=sc.nextInt();
        String []studentName= new String[num];

        //Take input from user
        System.out.println("Enter the elements: ");
        for(int i=0;i<studentName.length;i++){
            studentName[i]=sc.next();
        }
        //Print array elements
        for(int j =0;j<studentName.length;j++)
        {
            System.out.println("Index "+j+" element:
"+studentName[j]);
        }
        sc.close();
    }
}
```

### Output:-

```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3
Enter the size of array: 4
Enter the elements:
Ashlesha
Ruchika
Priti
Sai
Index 0 element: Ashlesha
Index 1 element: Ruchika
Index 2 element: Priti
Index 3 element: Sai

Process finished with exit code 0
|
```

### 3. Create Heterogenous Array.

Create parent class Employee having name, department

Create sub-class(child) Engineer class from Employee having incentive

Create another sub-class Manager from Employee having project

Perform Over-ride constructor in sub-class along with respective toString function as well.

#### Program:-

##### TestEmployee.java

```
package day2.heterogenousarray;

public class TestEmployee {
    public static void main(String[] args){
        Employee[] staff=new Employee[4];

        staff[0] =new Engineer("Ashlesha", 1 ,500);
        staff[1] =new Manager("Ruchika", 2 ,"IT");

        staff[0].calculateSalary();
        staff[1].calculateSalary();

        System.out.println(staff[0]);
        System.out.println(staff[1]);
        //System.out.println(staff[0].equals(staff[1]));

        //for each
        for(Employee e :staff){
            if(e instanceof Engineer){
                System.out.println("Engineer");
                Engineer engineer = (Engineer) e;
                System.out.println(engineer.getIncentives());
            }
            else if(e instanceof Manager) {
                System.out.println("Manager");
                Manager manager =(Manager) e;
                System.out.println(manager.getprojects());
            }
            else{
                System.out.println("not defined");
            }
        }
    }
}
```

## Employee.java

```
package day2.heterogenousarray;

public abstract class Employee {
    String name ;
    int id;
    float salary;

    public Employee (String name,int id){
        this.name= name;
        this.id = id;
    }

    public abstract void calculateSalary();
    @Override
    public String toString(){
        return name +" " + id +" ";
    }
    // @Override
    // public boolean equals(Object obj1){
    //     if(obj1 !=null && obj1 instanceof Employee){
    //         Employee e =(Employee) obj1;
    //         if (this.id == e.id && this.name.equals(e.name))
    //             return true;
    //     }
    //     return false;
    // }
}
```

## Engineer.java

```
package day2.heterogenousarray;

public class Engineer extends Employee {
    int salary;
    int incentives ;

    Engineer(String name ,int id,int incentives){
        super(name,id);
        this.incentives= incentives;
    }
    public int getIncentives(){
```

```

        return this.incentives;
    }

    @Override
    public void calculateSalary() {
        this.salary = 1000* id +incentives;
    }

    @Override
    public String toString() {
        return super.toString() + " " + this.salary+" " +
this.incentives;
    }
}

```

### Manager.java

```

package day2.heterogenousarray;

public class Manager extends Employee {
    String projects;

    Manager(String name ,int id,String projects){
        super(name,id);
        this.projects= projects;
    }

    public String getprojects(){
        return this.projects;
    }

    @Override
    public void calculateSalary() {
        this.salary = 5000 * id ;
    }

    @Override
    public String toString() {
        return super.toString() + " " + this.salary+" " +
this.projects;
    }
}

```



## Output:-

```
"C:\Program Files\Java\jdk-11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2
Ashlesha 1 1500 500
Ruchika 2 10000.0 IT
Engineer
500
Manager
IT
not defined
not defined

Process finished with exit code 0
```